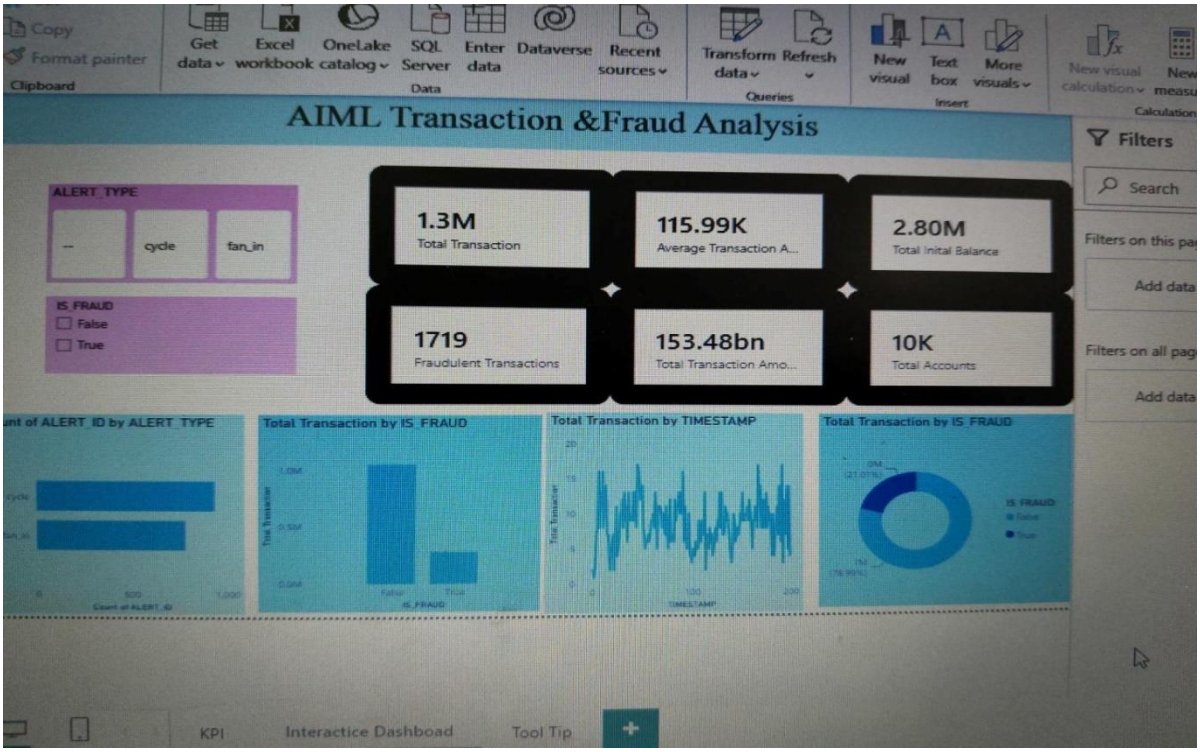
AIML Transaction & Fraud Analysis Dashboard



NAME	**G Ezhilarasi**
Course	**B.Com FinTech with AI**
Date	**19 October 2026**
Data source	**AIML Transaction & Fraud Analysis (Power BI report)**

Contents

No.	Section
1	Introduction
2	Dashboard Overview
3	KPI Analysis
4	Fraud Analysis
5	Alert Type Analysis
6	Transaction Trend over Time
7	Account and Balance Analysis
8	Tool Tip Page
9	Business Insights
10	Business Recommendations
11	Limitations and Data Notes
12	Conclusion

1. Introduction

1.1 About the dataset

The Global Sales and Financial Performance dataset analysed in this report is the AIML Transaction & Fraud Analysis dataset, presented in an interactive Power BI report. It records about 1.3 million financial transactions made across 10,000 accounts, with a total transaction amount of 153.48 billion. Each transaction carries a fraud flag (IS_FRAUD), and suspicious activity is classified by alert type (ALERT_TYPE) into two patterns, cycle and fan_in. The dashboard also tracks the total initial balance of the accounts (2.80 million) and the movement of transactions over time (TIMESTAMP).

1.2 Purpose of the dashboard

The dashboard gives a risk or finance team one place to monitor transaction volume, transaction value, fraud counts, alert patterns and account balances. Two slicers, ALERT_TYPE and IS_FRAUD, filter the visuals so that a user can isolate fraudulent or legitimate transactions, or a single alert pattern, without building a new report.

1.3 Purpose of this report

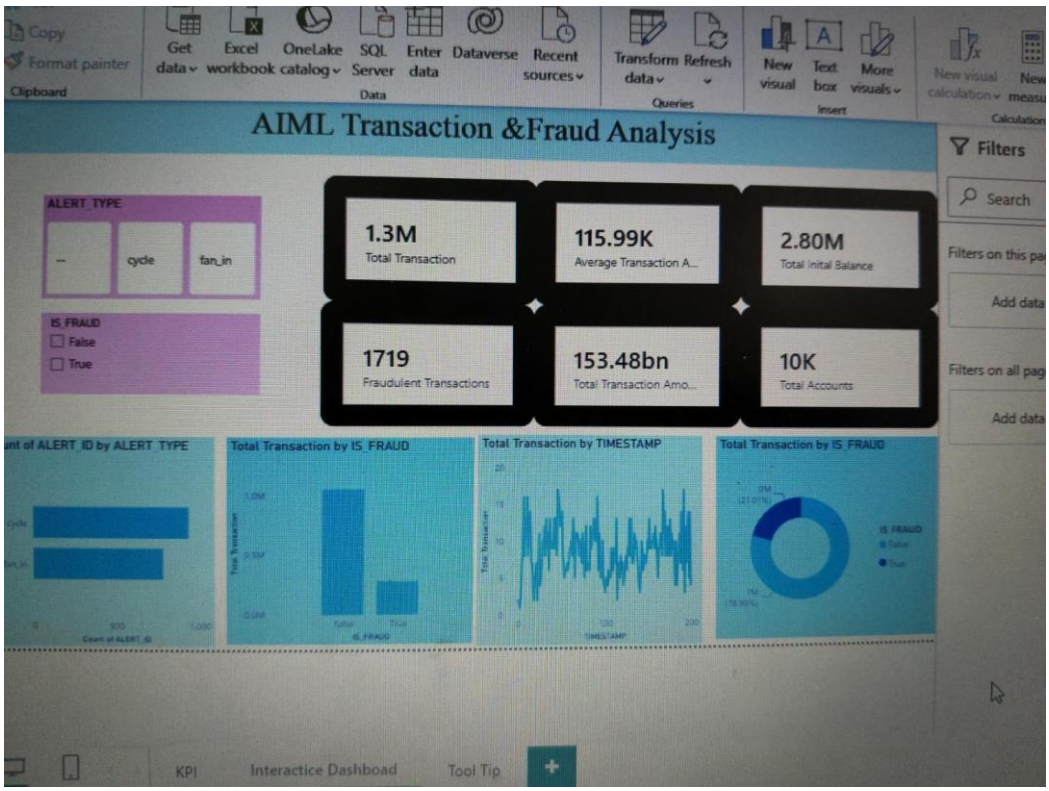
The dashboard shows what the data says. This report explains what it means. It interprets each KPI and chart, separates observations from insights, and turns them into practical recommendations for fraud monitoring, anti-money-laundering (AML) controls and data quality, which are core topics for FinTech with AI.

1.4 Objectives

- To summarise transaction volume, value, account and balance KPIs.
- To analyse fraud-flagged transactions and the alert types raised.
- To review the transaction pattern over time.
- To identify data-quality issues and recommend actions.

2. Dashboard Overview

2.1 Dashboard screenshot



Interactive Dashboard page: slicers, KPI cards and four charts

2.2 Pages in the report

The Power BI report has three pages, shown as tabs at the bottom of the screen: KPI, Interactive Dashboard and Tool Tip. The KPI page presents the six headline measures together with four charts. The Interactive Dashboard page adds the two slicers. The Tool Tip page is a small report page that appears as a pop-up when a user hovers over a visual.

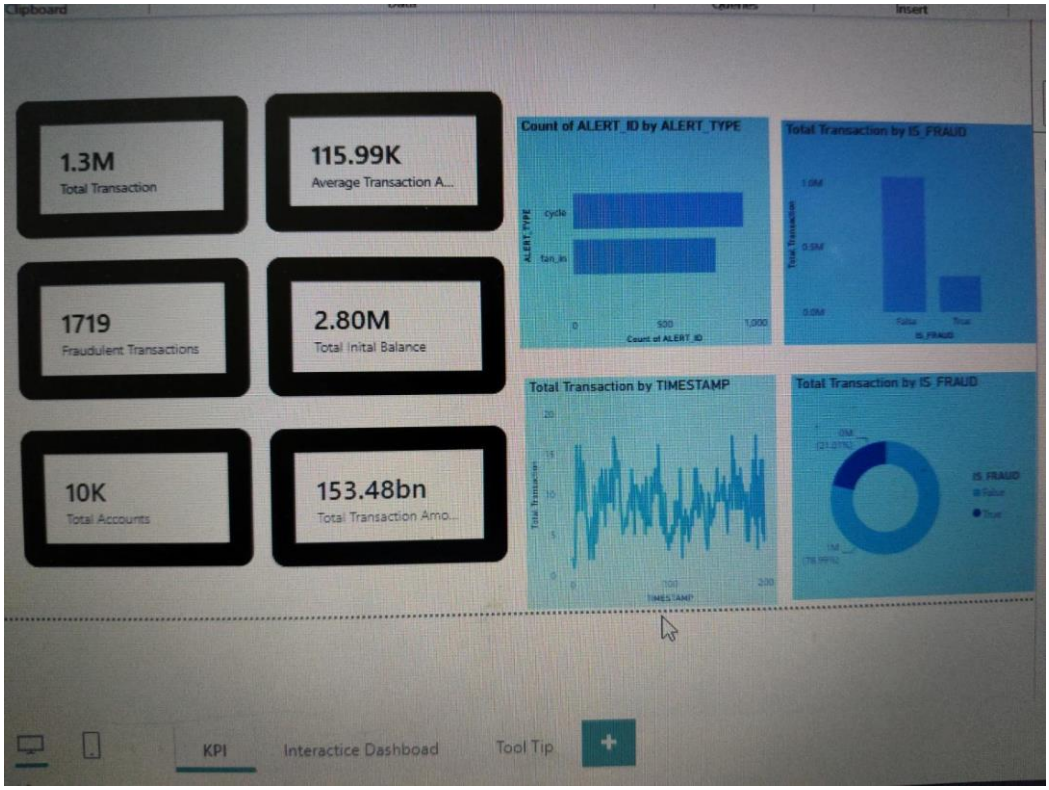
2.3 Key components

- KPI cards (six): Total Transaction, Average Transaction Amount, Total Initial Balance, Fraudulent Transactions, Total Transaction Amount and Total Accounts.
- Count of ALERT_ID by ALERT_TYPE: a horizontal bar chart comparing the two alert types, cycle and fan_in.
- Total Transaction by IS_FRAUD (column chart): total transactions where IS_FRAUD is False compared with True.
- Total Transaction by IS_FRAUD (donut chart): the percentage split of the same measure, 78.99% False and 21.01% True.
- Total Transaction by TIMESTAMP: a line chart of the transaction measure across the timestamp axis, from 0 to about 200.
- Slicers: ALERT_TYPE (including a blank option, cycle and fan_in) and IS_FRAUD (False and True).

2.4 Interactive filters

Selecting cycle or fan_in in the ALERT_TYPE slicer, or False or True in the IS_FRAUD slicer, re-draws every visual on the page. This allows a manager to check, for example, how many fraud-flagged transactions belong to cycle alerts.

3. KPI Analysis



KPI page: six KPI cards and four charts

Table 1. KPIs shown on the dashboard

KPI	Value	Meaning
Total Transaction	1.3M	Number of transactions recorded
Average Transaction Amount	115.99K	Mean value per transaction
Total Initial Balance	2.80M	Opening balances of all accounts
Fraudulent Transactions	1,719	Transactions flagged IS_FRAUD = True
Total Transaction Amount	153.48bn	Value of all transactions
Total Accounts	10K	Accounts in the dataset

3.1 Derived ratios

The ratios below are calculated from the printed KPI values. Because the KPI cards show rounded figures, the results are indicative.

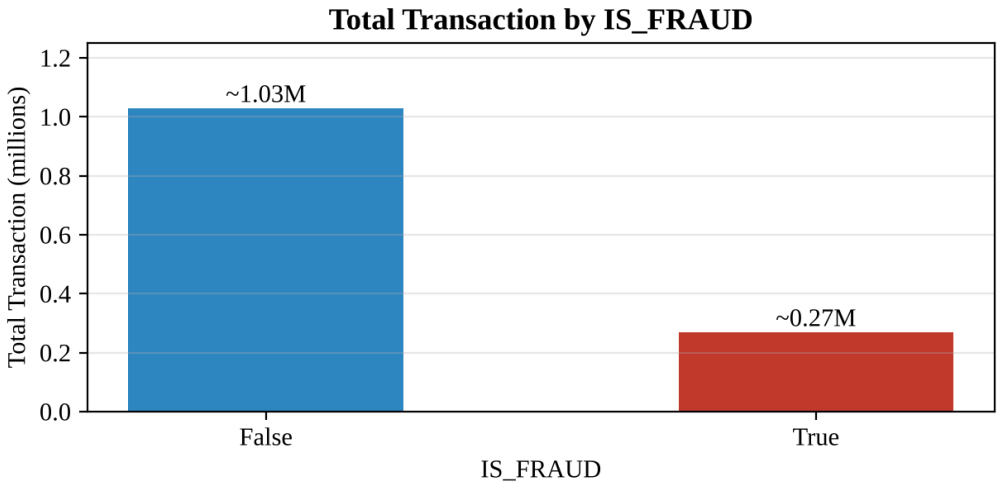
Table 2. Derived ratios

Ratio	Calculation	Result
Implied transaction count	153.48bn / 115.99K	about 1.32M
Fraud rate by count	1,719 / 1.32M	about 0.13%
Transactions per account	1.32M / 10K	about 132
Transaction amount per account	153.48bn / 10K	about 15.35M
Average initial balance per account	2.80M / 10K	about 280
Average transaction vs average opening balance	115.99K / 280	about 414 times

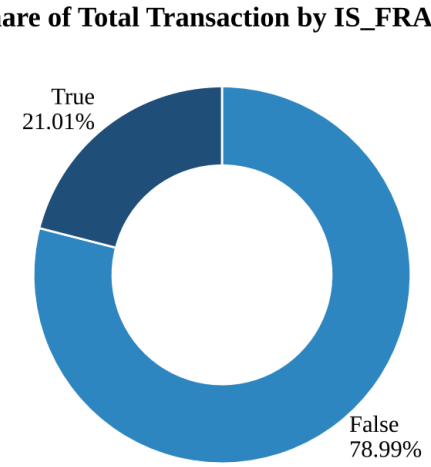
The implied count of about 1.32 million agrees with the 1.3M shown on the card, which is a useful consistency check. Fraud flagged by count is rare, at about 0.13 percent, yet each account handles about 132 transactions, so even a small fraud rate affects many accounts.

4. Fraud Analysis

4.1 Total Transaction by IS_FRAUD



Total Transaction by IS_FRAUD (values approximate, recreated from the dashboard)



Share of Total Transaction by IS_FRAUD

Table 3. Fraud indicators

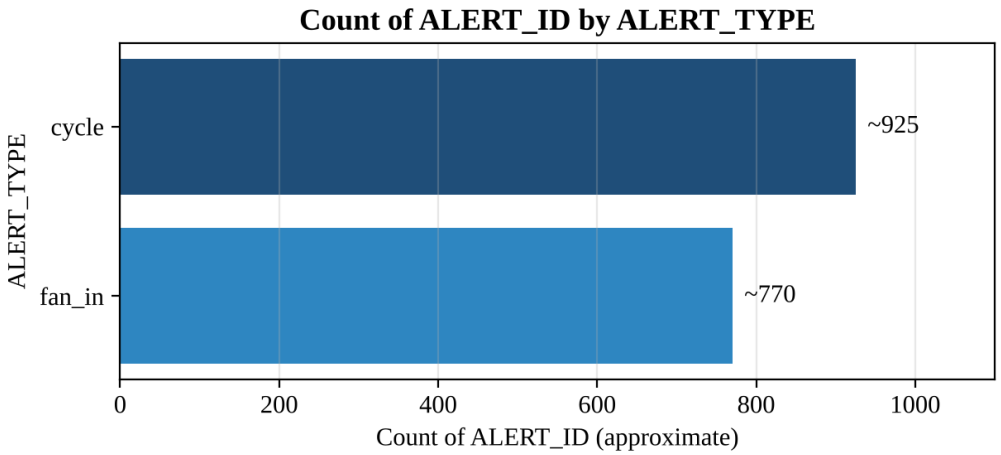
Indicator	Value	Observation
IS_FRAUD = False (column chart)	about 1.03M	Large majority of the measure
IS_FRAUD = True (column chart)	about 0.27M	Smaller bar, roughly a quarter of False
IS_FRAUD = False (donut)	78.99%	Share of Total Transaction
IS_FRAUD = True (donut)	21.01%	Share of Total Transaction
Fraudulent Transactions (KPI)	1,719	About 0.13% of transactions by count

The column and donut charts agree with each other: about 79 percent of the Total Transaction measure belongs to legitimate transactions and about 21 percent to fraud-flagged ones.

4.2 Metric reconciliation

The donut chart puts 21.01 percent of the Total Transaction measure under IS_FRAUD = True, which is about 0.27 million of 1.3 million. The KPI card, however, reports only 1,719 fraudulent transactions, about 0.13 percent. These two figures cannot both describe the fraud share by count. They appear to use different definitions or different columns. The measure behind the charts should be documented before fraud exposure is reported to management.

5. Alert Type Analysis



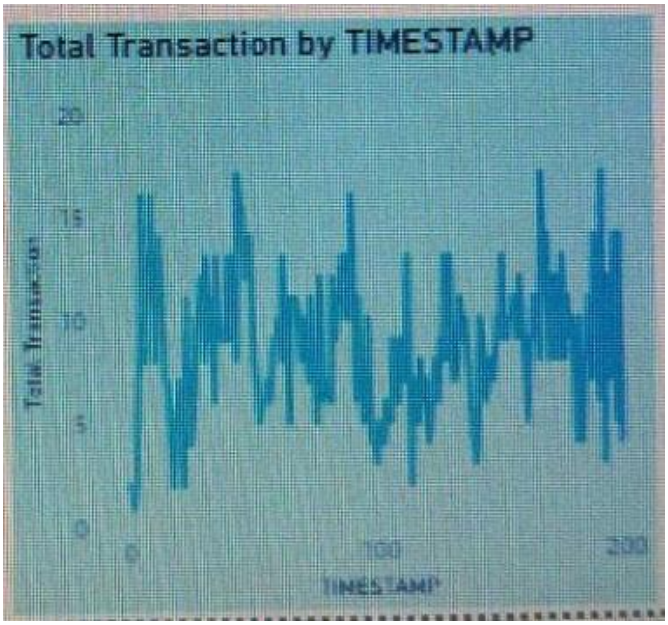
Count of ALERT_ID by ALERT_TYPE (values approximate)

Table 4. Alert types

Alert type	Approx. count	Pattern
cycle	about 925	Money moves through a loop of accounts and returns to its source
fan_in	about 770	Money from many accounts is collected into one account

Both alert types are well-known network patterns in AML monitoring. Cycle alerts are somewhat more frequent than fan_in alerts, about 925 against 770, or roughly 55 percent against 45 percent of the two bars combined. The two patterns are of similar size, so monitoring rules need to cover both.

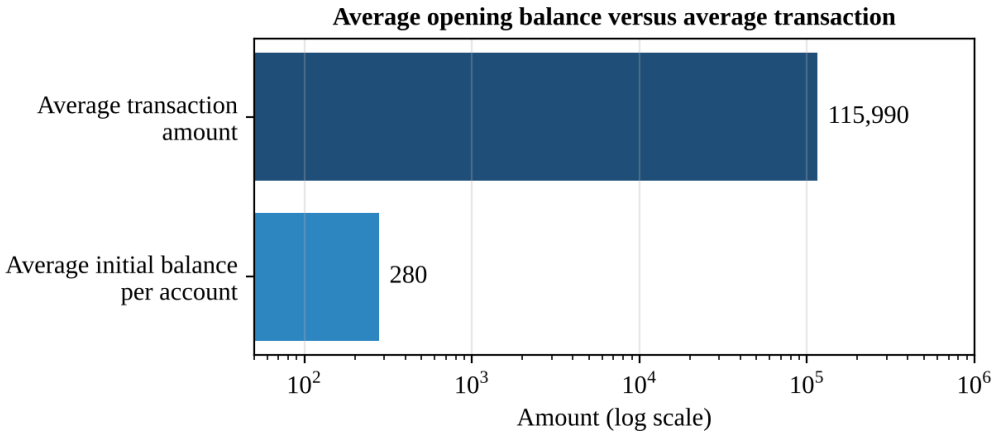
6. Transaction Trend over Time



Total Transaction by TIMESTAMP, as shown on the dashboard

The line chart fluctuates between about 0 and 20 on the vertical axis across a timestamp axis of 0 to about 200. The line moves up and down all the time and does not show one dominant spike or a steady trend. This points to a steady flow of transactions rather than a one-off event. The unit of the timestamp (for example day or hour) is not visible, so seasonality or time-of-day patterns cannot be confirmed from the screenshot.

7. Account and Balance Analysis

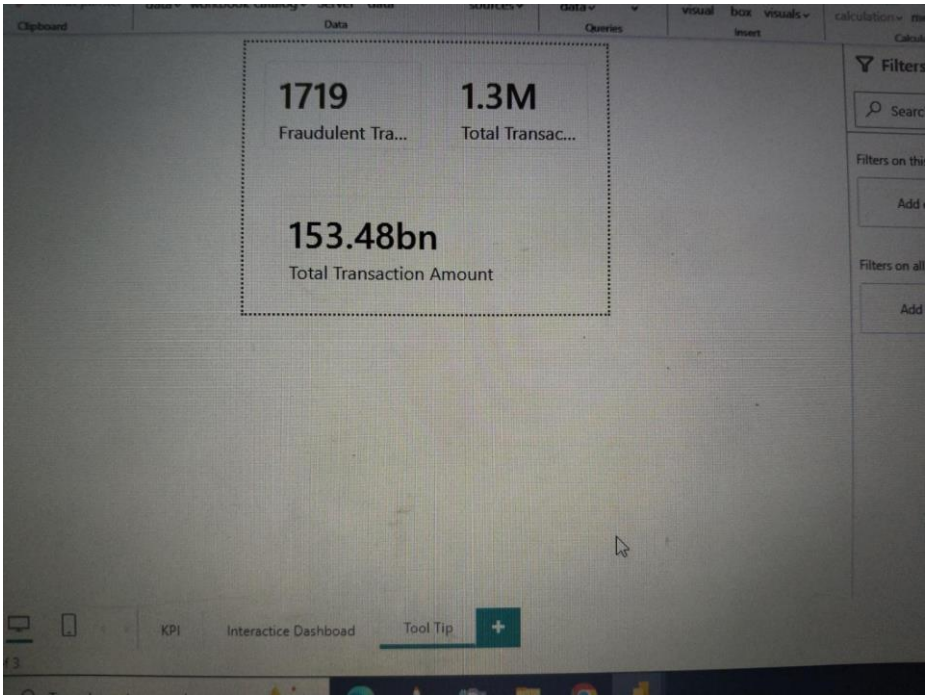


Average initial balance per account versus average transaction amount (log scale)

The 10,000 accounts hold a total initial balance of 2.80 million, or about 280 per account, but the average transaction is 115.99K, about 414 times larger. Total transaction value of 153.48 billion is roughly 55,000 times the total initial balance. In real banking data, transactions of this size would need a funding source far larger than the opening balances. This suggests that the data may be simulated, that balances are recorded differently from transaction flows, or

that money is being passed repeatedly between accounts, which is exactly what cycle alerts describe. Either way, the relationship should be reviewed.

8. Tool Tip Page



Tool Tip page

The Tool Tip page shows three values, 1719 Fraudulent Transactions, 1.3M Total Transactions and 153.48bn Total Transaction Amount. When a user hovers over a chart, this page can appear as a small pop-up, giving context without leaving the main page. Its figures match the KPI page.

9. Business Insights

Table 5. Observations and insights

Observation	Insight
1,719 fraudulent transactions out of about 1.3M (0.13%).	Fraud by count is rare, so detection needs precise rules and good alert prioritisation to avoid missing it among a very large volume.
Donut chart shows 21.01% of the measure as fraud, but the KPI shows 0.13% by count.	The fraud measure is defined inconsistently, or the two figures use different bases. Reporting both without explanation could mislead management.
Cycle and fan_in alerts are both frequent (about 925 and 770).	Layering and aggregation patterns are both present, so monitoring should cover circular flows and many-to-one collection.
Average transaction (115.99K) is about 414 times the average opening balance (280).	Balances and transaction flows do not reconcile. Data may be synthetic or funds are being recycled between accounts.
Transaction line fluctuates between 0 and 20 with no dominant spike.	Activity is steady, so unusual spikes would stand out and can be used as simple alert triggers.

10. Business Recommendations

- Fraud measure: define Fraudulent Transactions and the IS_FRAUD measure clearly, and report both the count-based and the value-based fraud rate.
- Alert monitoring: keep rules for both cycle and fan_in patterns and review the larger cycle group first.
- AI/ML models: use the IS_FRAUD label to train and test a fraud-detection model, and measure it with precision and recall because fraud is rare (about 0.13%).
- Balance checks: reconcile account balances with transaction flows, and investigate accounts with transactions far above their opening balance.
- Time analysis: add a readable date or time field so seasonal and time-of-day patterns can be tested.
- Dashboard design: add data labels to the charts, add a tool tip showing both rates, and document each measure.

11. Limitations and Data Notes

- All values come from photographs of the dashboard. KPI values are quoted as printed; chart values (alert counts, column heights, 1.03M and 0.27M) are visual estimates.
- KPI cards show rounded figures (for example 1.3M), so derived ratios are approximate.
- The timestamp unit and the exact definition of the Total Transaction measure are not shown.
- The currency is not stated.
- No fraud loss amount, customer, merchant or location data appears, so the analysis is limited to what is on the dashboard.

12. Conclusion

The AIML Transaction & Fraud Analysis dashboard gives a compact view of about 1.3 million transactions worth 153.48 billion across 10,000 accounts. Only 1,719 transactions are flagged as fraud, about 0.13 percent by count, and alerts follow two patterns, cycle and fan_in. The main issues are that the fraud figure on the KPI card does not match the 21.01 percent share in the chart, and that transaction values are far above account balances. Resolving these two points and strengthening AI-based monitoring for both alert patterns would make the dashboard a more reliable tool for fraud and AML decisions.

Source

AIML Transaction & Fraud Analysis Power BI dashboard (KPI, Interactive Dashboard and Tool Tip pages), supplied as photographs. Tables and charts were prepared by G Ezhilarasi from these sources.